
CV Model Lab Report

Showcase Demo
Model: Model A — Baseline

Generated	2026-06-03 10:49
IoU Threshold	0.50
Confidence Threshold	0.25
Class-aware Matching	Yes
Ignore Crowd	Yes

Executive Summary

Images 40	Ground Truth 87	Predictions 72	Images w/ Errors 18
TP 64	FP 8	FN 23	F1 80.5%
Precision 88.9%	Recall 73.6%	Health Errors 0	Health Warnings 1
Critical Recs 1	Warning Recs 4	Total Recs 6	Categories 5

Dataset summary

Metric	Value
Images	40
Annotations (GT)	87
Categories	5
Predictions (before threshold)	72
Predictions (after threshold)	72
Missing image files	0

Overall metrics

Metric	Value
TP	64
FP	8
FN	23
Precision	88.9%
Recall	73.6%
F1	80.5%
Micro Precision	88.9%
Micro Recall	73.6%
Micro F1	80.5%
Macro Precision	93.1%
Macro Recall	75.8%
Macro F1	82.1%

COCO AP Metrics

Metric	Value
AP@[.5:.95]	52.4%
AP50	74.1%
AP75	55.8%
APsmall	27.6%
APmedium	55.1%
APlarge	71.2%
AR1	38.1%
AR10	61.2%
AR100	63.7%

ARsmall	31.8%
ARmedium	64.1%
ARlarge	76.3%

Per-class AP (top 5)

Class	AP	AP50	AP75	AR
red_light	58.1%	80.2%	61.7%	68.1%
yellow_light	43.2%	63.5%	44.9%	55.8%
green_light	59.1%	81.2%	62.2%	69.5%
pedestrian_sign	51.4%	73.1%	54.0%	62.6%
background_distractor	50.1%	72.4%	52.8%	62.4%

Per-class metrics

Top 10 weakest classes by Recall

Class	GT	Pred	TP	FP	FN	Precision	Recall	F1
background_distractor	20	10	10	0	10	100.0%	50.0%	66.7%
yellow_light	12	8	8	0	4	100.0%	66.7%	80.0%
green_light	25	24	19	5	6	79.2%	76.0%	77.6%
red_light	22	22	19	3	3	86.4%	86.4%	86.4%
pedestrian_sign	8	8	8	0	0	100.0%	100.0%	100.0%

Top 10 classes by FP count

Class	GT	Pred	TP	FP	FN	Precision	Recall	F1
green_light	25	24	19	5	6	79.2%	76.0%	77.6%
red_light	22	22	19	3	3	86.4%	86.4%	86.4%
yellow_light	12	8	8	0	4	100.0%	66.7%	80.0%
pedestrian_sign	8	8	8	0	0	100.0%	100.0%	100.0%
background_distractor	20	10	10	0	10	100.0%	50.0%	66.7%

Top 10 classes by FN count

Class	GT	Pred	TP	FP	FN	Precision	Recall	F1
background_distractor	20	10	10	0	10	100.0%	50.0%	66.7%
green_light	25	24	19	5	6	79.2%	76.0%	77.6%
yellow_light	12	8	8	0	4	100.0%	66.7%	80.0%
red_light	22	22	19	3	3	86.4%	86.4%	86.4%
pedestrian_sign	8	8	8	0	0	100.0%	100.0%	100.0%

Dataset health check

Errors 0	Warnings 1	Info 6	Missing Images 0
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Top 7 issues

Severity	Type	File	Message
Info	imageWithoutGroundTruth	scene_029.png	Image "scene_029.png" has no annotations.
Info	imageWithoutGroundTruth	scene_030.png	Image "scene_030.png" has no annotations.
Info	imageWithoutGroundTruth	scene_031.png	Image "scene_031.png" has no annotations.
Info	imageWithoutGroundTruth	scene_032.png	Image "scene_032.png" has no annotations.
Info	imageWithoutGroundTruth	scene_033.png	Image "scene_033.png" has no annotations.
Info	predictionOnImageWithoutGroundTruth	scene_032.png	An image has predictions but no ground truth.
Warning	rareClass		Class "pedestrian_sign" has few GT objects.

Worst cases

Most Errors

File	Reason	TP	FP	FN	Score	IoU
scene_023.png	1 FP + 3 FN	2	1	3		
scene_016.png	1 FP + 2 FN	1	1	2		
scene_020.png	1 FP + 2 FN	2	1	2		
scene_009.png	1 FP + 1 FN	1	1	1		
scene_022.png	0 FP + 2 FN	1	0	2		
scene_026.png	0 FP + 2 FN	2	0	2		
scene_032.png	2 FP + 0 FN	0	2	0		
scene_038.png	1 FP + 1 FN	3	1	1		
scene_040.png	0 FP + 2 FN	2	0	2		
scene_001.png	0 FP + 1 FN	0	0	1		

High Confidence FP

File	Reason	TP	FP	FN	Score	IoU
scene_023.png	FP at score 0.84	2	1	3	0.844	

Most False Negatives

File	Reason	TP	FP	FN	Score	IoU
scene_023.png	3 missed objects	2	1	3		
scene_016.png	2 missed objects	1	1	2		
scene_020.png	2 missed objects	2	1	2		
scene_022.png	2 missed objects	1	0	2		
scene_026.png	2 missed objects	2	0	2		
scene_040.png	2 missed objects	2	0	2		
scene_001.png	1 missed objects	0	0	1		
scene_005.png	1 missed objects	0	0	1		
scene_009.png	1 missed objects	1	1	1		
scene_010.png	1 missed objects	1	0	1		

Class Confusion

File	Reason	TP	FP	FN	Score	IoU
scene_009.png	1 confused object(s)	1	1	1		
scene_016.png	1 confused object(s)	1	1	2		
scene_019.png	1 confused object(s)	2	1	0		
scene_020.png	1 confused object(s)	2	1	2		
scene_023.png	1 confused object(s)	2	1	3		
scene_038.png	1 confused object(s)	3	1	1		

Small Missed Objects

File	Reason	TP	FP	FN	Score	IoU
scene_016.png	2 small object(s) missed	1	1	2		
scene_022.png	2 small object(s) missed	1	0	2		
scene_017.png	1 small object(s) missed	1	0	1		
scene_018.png	1 small object(s) missed	2	0	1		
scene_020.png	1 small object(s) missed	2	1	2		
scene_021.png	1 small object(s) missed	2	0	1		
scene_038.png	1 small object(s) missed	3	1	1		
scene_040.png	1 small object(s) missed	2	0	2		

Confusion matrix

Top confused pairs (error cells only)

GT Class	Pred Class	Count	Row %
background_distractor	__missed__	8	40.0%
green_light	__missed__	6	24.0%
red_light	__missed__	3	13.6%
__background_fp__	green_light	2	50.0%
__background_fp__	red_light	2	50.0%
yellow_light	__missed__	2	16.7%
yellow_light	green_light	2	16.7%
background_distractor	green_light	1	5.0%
background_distractor	red_light	1	5.0%

Model Comparison

Leaderboard

#	Model	P	R	F1	TP	FP	FN	AP	AP50
1	Model C — High Precision	100.0%	80.5%	89.2%	70	0	17	61.2%	82.2%
2	Model A — Baseline	88.9%	73.6%	80.5%	64	8	23	52.4%	74.1%
3	Model B — High Recall	67.9%	82.8%	74.6%	72	34	15	47.1%	69.5%

Per-Class Ranking

Class	Best model	Worst model	F1 spread
yellow_light	Model C — High Precision	Model B — High Recall	46.7%
background_distractor	Model B — High Recall	Model C — High Precision	37.9%
green_light	Model C — High Precision	Model B — High Recall	28.0%
pedestrian_sign	Model A — Baseline	Model B — High Recall	23.1%
red_light	Model C — High Precision	Model B — High Recall	18.2%

Recommendations

Critical (1)

Title	Message	Action
Many missed objects	False negatives are a major part of the current error profile.	Inspect FN cases, annotation consistency, input resolution, augment...

Warnings (4)

Title	Message	Action
Class confusion: "yellow_light" predi...	The model confuses these two classes in class-agnostic matching.	Review annotation rules and visual similarity. Add discriminative e...
Class confusion: "background_distract...	The model confuses these two classes in class-agnostic matching.	Review annotation rules and visual similarity. Add discriminative e...
Class confusion: "background_distract...	The model confuses these two classes in class-agnostic matching.	Review annotation rules and visual similarity. Add discriminative e...
High-confidence false positives	Some false positives have high confidence scores, which makes thres...	Inspect hard negatives, add background examples, check score calibr...

Info (1)

Title	Message	Action
Rare class "pedestrian_sign"	This class has too few ground-truth examples.	Collect more samples or avoid trusting metrics for this class.

Appendix: Top Error Examples

File	Category	Type	Score	IoU
scene_001.png	red_light	FN		

scene_005.png	yellow_light	FN		
scene_009.png	yellow_light	FN		
scene_009.png	green_light	FP	0.592	
scene_010.png	red_light	FN		
scene_016.png	red_light	FP	0.491	
scene_016.png	background_distractor	FN		
scene_016.png	background_distractor	FN		
scene_017.png	background_distractor	FN		
scene_018.png	background_distractor	FN		
scene_019.png	green_light	FP	0.522	
scene_020.png	red_light	FP	0.568	
scene_020.png	green_light	FN		
scene_020.png	background_distractor	FN		
scene_021.png	background_distractor	FN		
scene_022.png	background_distractor	FN		
scene_022.png	background_distractor	FN		
scene_023.png	yellow_light	FN		
scene_023.png	green_light	FP	0.844	
scene_023.png	green_light	FN		
scene_023.png	green_light	FN		
scene_024.png	red_light	FN		
scene_026.png	yellow_light	FN		
scene_026.png	green_light	FN		
scene_032.png	red_light	FP	0.594	
scene_032.png	green_light	FP	0.556	
scene_034.png	green_light	FN		
scene_038.png	green_light	FP	0.541	0.002
scene_038.png	background_distractor	FN		
scene_040.png	green_light	FN		
scene_040.png	background_distractor	FN		